Contrast Enhancement by Histogram Warping

Digital Image Processing Project

Group 4

Digital Image Processing

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Introduction

Contrast enhancement is a fundamental task in digital image processing for improving the visual quality of images. Enhanced contrast improves picture interpretation in applications such as remote sensing and medical imaging. Many real-world photographs have multimodal histograms, with peaks representing diverse regions such as shadows, background, and highlights. Preserving these modes is critical to sustaining the image's perceptual structure. However, for real-world images with multimodal histograms, histogram equalization performs poorly since it may lead to excessive amplification of certain intensity regions and loss of significant histogram modes, resulting in unnatural visual artifacts [1]. Automatic contrast enhancement by histogram warping is designed for grayscale images with multimodal histograms. The technique improves the contrast between the features by widening the modes of an image histogram [1].

1 Histogram Warping

Histogram warping is a global contrast technique that modifies intensity values based on the identified histogram structure instead of employing a uniform distribution. It improves feature contrast by widening or pushing the modes of an image histogram farther apart while keeping the original order. Grundland and Dodgson developed the method to address the limitations of classic histogram equalization [1].

The primary step is to identify the important modes (peaks or regions of high pixel concentration) and valleys (low frequency regions) of the image histogram. To find these modes and valleys, the histogram is first smoothed to remove noise and minor fluctuations, as raw histograms contain noise and small fluctuations that can lead to inaccurate detection of relevant features. Histogram smoothing preserves the distribution's overall form while removing unimportant fluctuations. Once the significant valleys have been identified, the intensity range is divided into segments defined by adjacent valleys. Unlike histogram equalization, which redistributes all intensity values uniformly, histogram warping addresses each segment individually. For each segment, parameters are defined to control how the intensity values are redistributed in the output image.

A monotonic transformation function is then constructed to remap the original intensity values. This transformation increases the contrast by pushing histogram modes farther apart, widening the gap between prominent histogram modes, and evenly distributing them across the available dynamic range. Monotonicity guarantees that the ordering of pixel intensities is preserved, preventing artifacts such as intensity reversal or false edges [1]. Histogram warping preserves multimodal qualities by distributing intensity values based on recognized histogram features or structures, resulting in visually natural results.

2 Methodology

The method enhances image contrast by analysing and modifying the structure of the image histogram through smoothing, segmentation and monotonic intensity transformation.

2.1 Histogram Computation and Smoothing

Let the input image be a grayscale image with T discrete gray levels. The histogram is defined as:

$$h = [h_0, h_1, \ldots, h_{T-1}] \tag{1}$$

where h_i represents the number of pixels with intensity level i.

The smoothed estimate of the histogram is computed using a regularization-based smoothness priors approach. The smoothed histogram f is obtained by minimization problem

$$\min_f \left(\|f - h\|^2 + \lambda^2 \|D_2 f\|^2 \right) \tag{2}$$

where D_2 is a second-order difference operator enforcing smoothness and $\lambda > 0$ is regularization parameter. This is a classical inverse problem with regularization parameter and the solution for the minimization problem is

$$\hat{f} = \left(I_T + \lambda^2 D_2^T D_2 \right)^{-1} h, \tag{3}$$

where I_T is identity matrix of size $T \times T$.

2.2 Detection of Histogram Valleys and Segmentation

Valleys are identified by analysing the first derivative of the smoothed histogram f. Valley positions indicate local minima and divide the histogram into $K + 1$ segments.

2.3 Wrapping Parameters and Intensity Transformation Function

Histogram valleys are used for segmenting object features. The segments are simply defined as the spaces between histogram valleys and are treated as object features. For each segment, parameters a_k, b_k, and d_k control the shape and extent of the mapping. The intensity transformation function $T(x)$ is defined as a piecewise rational quadratic, one time differentiable, interpolating monotonic spline. The transformation function is derived by Grundland and Dodgson like so

$$\begin{aligned} T(x) &= b_{k-1} + \frac{r_k t^2 + d_{k-1}(1-t)t}{r_k + (d_k + d_{k-1} - 2r_k)(1-t)t}(b_k - b_{k-1}), \\ r_k &= \frac{b_k - b_{k-1}}{a_k - a_{k-1}} \text{ and } t = \frac{x - a_{k-1}}{a_k - a_{k-1}}, \ x \in [a_{k-1}, a_k]. \end{aligned} \tag{4}$$

Parameters a_k, b_k and d_k are initially defined with the transformation function, so that $b_k = T(a_k)$ and $d_k = T'(a_k)$. Grundland and Dodgson defines a_k as the midpoint between histogram valleys $a_k = (v_{k-1} + v_k)/2$ and added histogram equalization only in the midpoint of the object feature to enhance the contrast of the object features between the histogram valleys [1]. The mapping widens dominant histogram modes, increasing contrast between distinct image regions.

3 Results and Findings

The original and improved images, as well as the smoothed histogram before and after warping, were plotted for each image. By eliminating noise and tiny variations, the smoothed histogram facilitates the identification of the prominent modes and troughs. After the transformation, the modes are pushed farther apart, as seen by the wraped histogram.

The parameter λ was given in the assignment and is used to limit distortion in the slope values d_k, ensuring that the intensity mapping remains monotonic and does not introduce artifacts. The additional parameters τ and δ were chosen experimentally. The value $\tau = 0.01$ was used to ignore unreliable histogram behavior near the very low and very high intensity ends, which prevents false valley detection at the boundaries. The parameter $\delta = 0.02$ controls how finely the CDF is segmented; smaller δ values produce stronger enhancement while larger values produce weaker enhancement. The chosen settings gave stable contrast enhancement on all test images.

Histogram warping improved contrast by separating the major modes of the histogram, which made details in the images more visible (e.g., clouds, textures, buildings). The results look natural because the method preserves the order of intensities and does not flatten the histogram. Compared to histogram stretching, warping is more adaptive because it uses histogram structure rather than applying a global linear change. Compared to histogram equalization, warping avoids over-enhancement and noise amplification, so the images look more realistic.

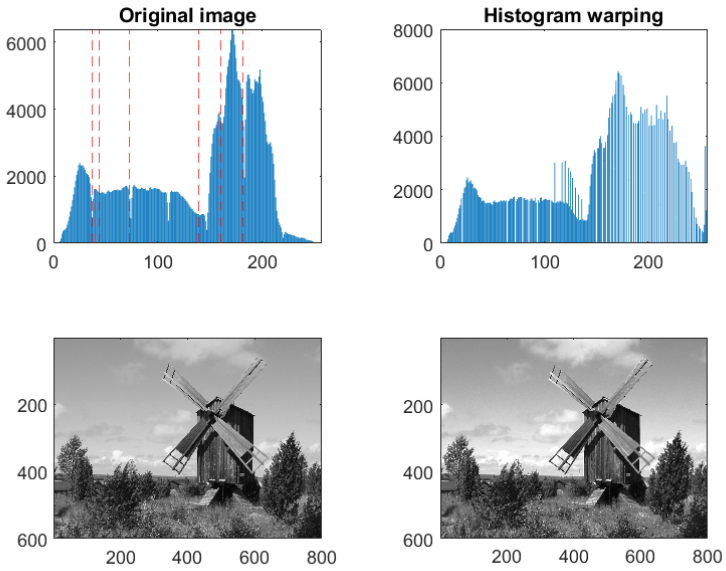


Figure 1: Windmill: original and warped results with histograms. Red dashed lines on the histogram of the original image are the histogram valleys used in the histogram warping.

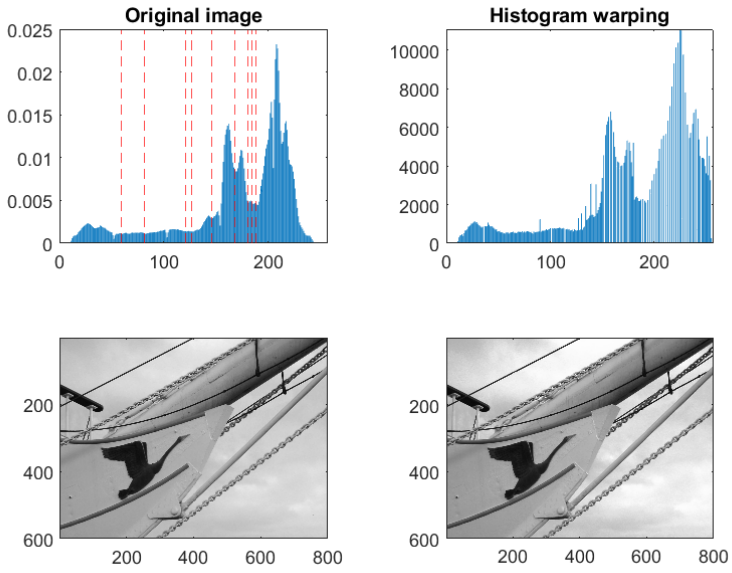


Figure 2: Swan: original and warped results with histograms. Red dashed lines on the histogram of the original image are the histogram valleys used in the histogram warping.

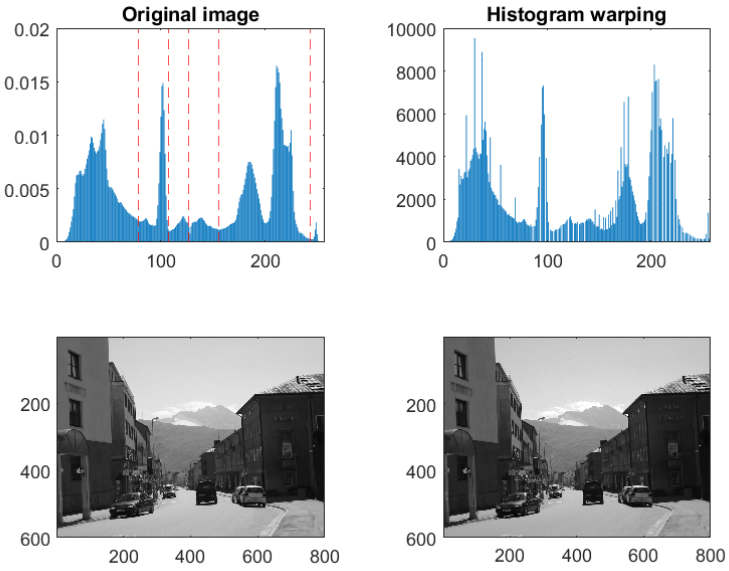


Figure 3: Ankenes: original and warped results with histograms. Red dashed lines on the histogram of the original image are the histogram valleys used in the histogram warping.

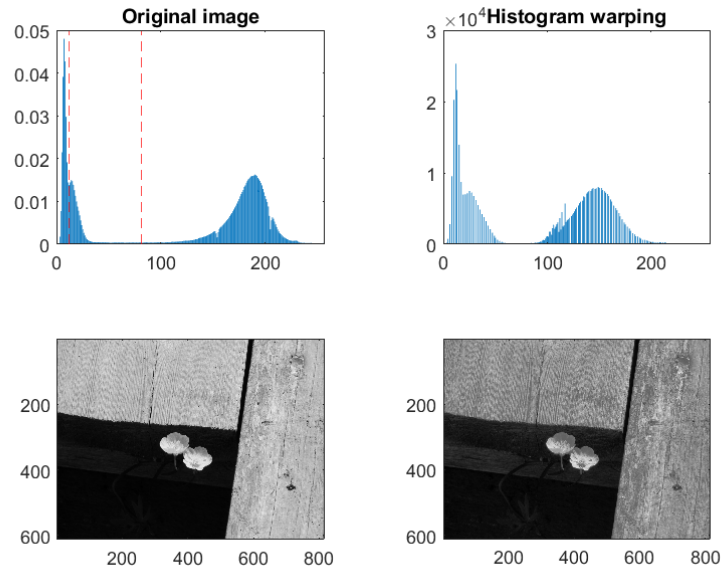


Figure 4: Leinikki: original and warped results with histograms. Red dashed lines on the histogram of the original image are the histogram valleys used in the histogram warping.

Conclusion

Automatic contrast enhancement using histogram wrapping technique results in overall visual quality of grayscale images with multi modal histograms. The traditional histogram equalization were inferior as they were prone to over amply noise and flatten tonal structures. Whereas the histogram warping offers better result by selectively expanding the meaningful intensity modes without changing the overall luminance ordering. The resulting images will have a contrast enhancement of natural looking without the need of introducing artifacts.

Experimental results confirms the ability of histogram warping to identity important histogram modes and increase the separation between, thereby enhancing the structural details in diverse scenes. A flexible control over contrast strength is possible with the smoothing parameter λ and warping coefficients a_k, b_k, d_k, allowing it to be applied over different image contents. For images containing multiple tonal modes, histogram warping is a robust and visually consistent alternative to global equalisation.

The method can be further improved in future extensions by including the optimization of valley selection, adaptive parameter tuning, and extending the method to color images with perceptual color spaces.

References

[1] Mark Grundland and Neil A. Dodgson. *AUTOMATIC CONTRAST ENHANCEMENT BY HISTOGRAM WARPING*, pages 293–300. Springer Netherlands, Dordrecht, 2006.